Fiscal Year:	FY 2019	**Task Last Updated:**	FY 11/06/2018
PI Name:	Rosi, Susanna Ph.D.		
Project Title:	The Role of Oxidative Stress and Inflammation on Synaptic Functions After Exposure to Space Radiation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rosis@ptrehab.ucsf.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	415-206-3708
Organization Name:	University of California San Francisco		
PI Address 1:	Physical Therapy and Neurological Surgery		
PI Address 2:	1001 Potrero Ave. Bldg #1 Room 101		
PI Web Page:			
City:	San Francisco	**State:**	CA
Zip Code:	94110-3518	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Space Radiobiology NNJ13ZSA001N
Start Date:	01/06/2014	**End Date:**	01/05/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 1/5/2020 per NSSC information (Ed., 2/12/19) NOTE: End date changed to 1/5/2019 per S. Monk/HRP-LaRC (Ed., 12/21/17)		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Huang, Ting-Ting Ph.D. (Palo Alto Institute for Research and Education, Inc.) Nelson, Gregory Ph.D. (Loma Linda University)		
Grant/Contract No.:	NNX14AC94G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Radiation exposure significantly affects the hippocampal formation, a structure critical for cognitive function. We demonstrated that low doses of 56Fe irradiation induced changes in cognition that correlated with changes in the expression of memory-related proteins and that low doses of space radiation and oxidative stress were associated with reductions in dendritic complexity and spine density. The underlying mechanisms for radiation-induced cognitive impairments are unknown, but likely are multifaceted, involving inflammation and chronic oxidative stress. Persistent oxidative stress has been demonstrated in space radiation studies in animal models; our preliminary data demonstrate that low doses of space radiation alter the expression of inflammatory chemokines and the dendritic system in the hippocampus. Maintenance of the dendritic system, including spine density and dendritic arbors, is essential for proper synaptic functions. Therefore, the dendritic system may be the missing link between oxidative stress/neuroinflammation and cognitive decline following irradiation. The purpose of this application is to determine how space-relevant doses of radiation affect dendritic maintenance as it pertains to synaptic functions and memory. We hypothesize that space radiation, even at low doses, impairs synaptic functions with negative consequences for cognition and that these effects are mediated by oxidative stress and inflammation. We will use novel mouse models with neuronal-specific overexpression of an antioxidant enzyme to address the role of oxidative stress. The mice will be exposed to space-relevant doses of proton, Helium, Oxygen, or a combination of the three and analyzed for late effects. Experimental endpoints will include cognitive functions, dendritic structures and synaptic functions, molecules that control the dendritic system, inflammatory responses, and indices of oxidative stress. This systematic approach will allow us: 1) to connect the functional output of cognition to the molecular pathway that controls synaptic plasticity and memory; and 2) to determine the role of oxidative stress and inflammation in space radiation-mediated changes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed aims directly address NASA's Human Research Program (HRP) Integrated Research Plan (IRP) program elements that specify a number of gaps related to the acute and late risks to the central nervous system (CNS). Specific gaps addressed by this proposal are the acute functional changes and the long-term CNS risk and the mechanisms of changes. Data obtained from our mouse models will also help to estimate late CNS risks to astronauts. On a broader scope, the use of mouse models to address the role of oxidative stress and inflammation in space radiation-mediated CNS risk should be beneficial in filling in the gaps of degenerative risks to the extent they apply to the CNS.
Task Progress:	We find chronic deficits in memory associated with whole body helium radiation as well as combination of Helium Protons and Oxygen. For the first time we show that transient treatment with a microglia depletion agent, one week after helium exposure is able to completely prevent chronic memory deficits, more than 90+ days after helium exposure. Moreover, we found that memory-deficit recovery was in part due to a reduced neuroinflammatory milieu and alterations in the repopulated microglia phenotype. These results demonstrate that even very transient microglia depletion up to a week after exposure could serve as a powerful therapeutic option for prevention of radiation-induced memory deficits during space missions. Furthermore, we investigated the long-term effect of galactic cosmic radiation (GCR) exposure on cognitive functions in male and female mice. Strikingly we found that male mice exposed to GCR simulation were impaired in many cognitive domains while females were not. The cognitive impairments observed in male mice were also paralleled by changes in synapse composition and microglia phenotype.
Bibliography Type:	Description: (Last Updated: 09/04/2023)
Articles in Peer-reviewed Journals	Krukowski K, Jones T, Campbell-Beachler M, Nelson G, Rosi S. "Peripheral T cells as a biomarker for oxygen-ion-radiation-induced social impairments." Radiat Res. 2018 Aug;190(2):186-93. Epub 2018 May 29. https://doi.org/10.1667/RR15046.1 ; PMID:29809107 , Aug-2018
Articles in Peer-reviewed Journals	Krukowski K, Feng X, Paladini MS, Chou A, Sacramento K, Grue K, Riparip LK, Jones T, Campbell-Beachler M, Nelson G, Rosi S. "Temporary microglia-depletion after cosmic radiation modifies phagocytic activity and prevents cognitive deficits." Sci Rep. 2018 May 18;8(1):7857. https://doi.org/10.1038/s41598-018-26039-7 ; PubMed PMID: 29777152; PubMed Central PMCID: PMC5959907 [Erratum in: Sci Rep. 2018 Jul 3;8(1):10297. PMID: 29967490] , May-2018
Articles in Peer-reviewed Journals	Krukowski K, Grue K, Frias ES, Pietrykowski J, Jones T, Nelson G, Rosi S. "Female mice are protected from space radiation-induced maladaptive responses." Brain Behav Immun. Brain Behav Immun. 2018 Nov;74:106-20. Epub 2018 Aug 11. https://doi.org/10.1016/j.bbi.2018.08.008 ; PubMed PMID: 30107198 , Nov-2018
Articles in Peer-reviewed Journals	Cekanaviciute E, Rosi S, Costes SV. "Central nervous system responses to simulated galactic cosmic rays." Int J Mol Sci. 2018 Nov 20;19(11):E3669. Review. https://doi.org/10.3390/ijms19113669 ; PMID: 30463349; PMCID: PMC6275046 , Nov-2018
Articles in Peer-reviewed Journals	Rosi S. "The final frontier: Transient microglia reduction after cosmic radiation exposure mitigates cognitive impairments and modulates phagocytic activity." Brain Circ. 2018 Jul-Sep;4(3):109-13. Review. https://doi.org/10.4103/bc.bc_24_18 ; PMID: 30450416; PMCID: PMC6187945 , Jul-2018